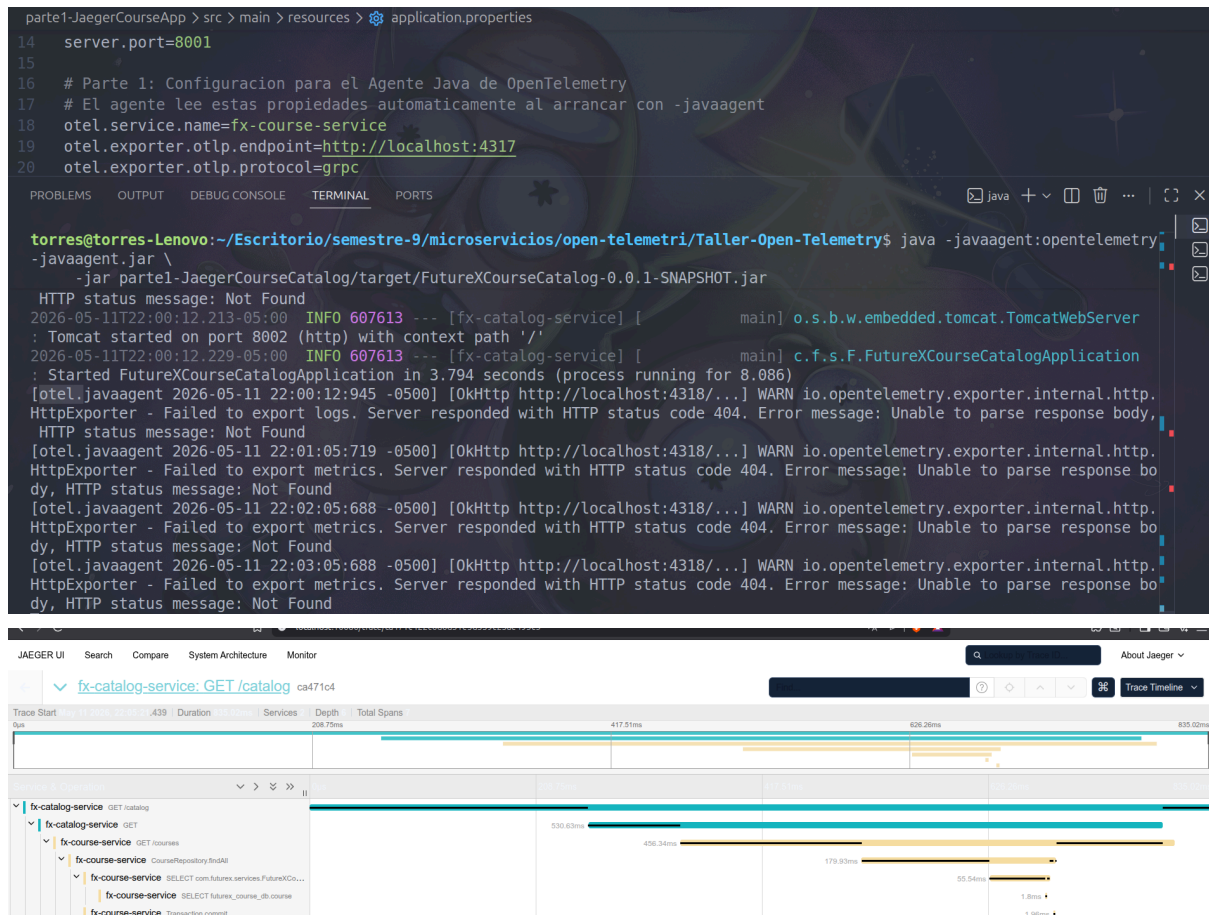


Alejandro Amu
Juan David Calderón
David Henao
Alejandro Torres

Parte 1 - Agente Java

Se muestra la configuración del agente y la consola mostrando el **OTel** activo.



```
parte1-JaegerCourseApp > src > main > resources > application.properties
14  server.port=8001
15
16  # Parte 1: Configuración para el Agente Java de OpenTelemetry
17  # El agente lee estas propiedades automáticamente al arrancar con -javaagent
18  otel.service.name=fx-course-service
19  otel.exporter.otlp.endpoint=http://localhost:4317
20  otel.exporter.otlp.protocol=grpc

torres@torres-Lenovo:~/Escritorio/semestre-9/microservicios/open-telemetry/Taller-Open-Telemetry$ java -javaagent:opentelemetry-
-javaagent.jar \
-jar parte1-JaegerCourseCatalog/target/FutureXCourseCatalog-0.0.1-SNAPSHOT.jar
HTTP status message: Not Found
2026-05-11T22:00:12.213-05:00 INFO 607613 --- [fx-catalog-service] [      main] o.s.b.w.embedded.tomcat.TomcatWebServer
: Tomcat started on port 8002 (http) with context path '/'
2026-05-11T22:00:12.229-05:00 INFO 607613 --- [fx-catalog-service] [      main] c.f.s.F.FutureXCourseCatalogApplication
: Started FutureXCourseCatalogApplication in 3.794 seconds (process running for 8.086)
[otel.javaagent 2026-05-11 22:00:12.945 -0500] [OkHttp http://localhost:4318/...] WARN io.opentelemetry.exporter.internal.http
HttpExporter - Failed to export logs. Server responded with HTTP status code 404. Error message: Unable to parse response body,
HTTP status message: Not Found
[otel.javaagent 2026-05-11 22:01:05.719 -0500] [OkHttp http://localhost:4318/...] WARN io.opentelemetry.exporter.internal.http
HttpExporter - Failed to export metrics. Server responded with HTTP status code 404. Error message: Unable to parse response bo
dy, HTTP status message: Not Found
[otel.javaagent 2026-05-11 22:02:05.688 -0500] [OkHttp http://localhost:4318/...] WARN io.opentelemetry.exporter.internal.http
HttpExporter - Failed to export metrics. Server responded with HTTP status code 404. Error message: Unable to parse response bo
dy, HTTP status message: Not Found
[otel.javaagent 2026-05-11 22:03:05.688 -0500] [OkHttp http://localhost:4318/...] WARN io.opentelemetry.exporter.internal.http
HttpExporter - Failed to export metrics. Server responded with HTTP status code 404. Error message: Unable to parse response bo
dy, HTTP status message: Not Found
```

The Jaeger UI screenshot shows a trace for the 'fx-catalog-service' with a duration of 417.51ms. The trace includes several spans, with the longest being 'fx-catalog-service GET /catalog' (417.51ms). The trace also shows a 'fx-course-service' span with a duration of 179.93ms.

Parte 2 - API Manual

Se muestra la configuración de Open Telemetry.

```

parte0-JaegerCourseApp > src > main > java > com > futurex > services > FutureXCourseApp > OpenTelemetryConfig.java > {} com.futurex.services.FutureXCourseApp
18 @Configuration
19 public class OpenTelemetryConfig {
20
21     @Value("${spring.application.name}")
22     private String applicationName;
23
24     @Value("${otel.exporter.otlp.endpoint:http://localhost:4317}")
25     private String otlpEndpoint;
26
27     @Bean
28     public OpenTelemetry openTelemetry() {
29         Resource resource = Resource.getDefault()
30             .merge(Resource.create(Attributes.of(AttributeKey.stringKey("service.name"), applicationName)));
31
32         OtlpGrpcSpanExporter spanExporter = OtlpGrpcSpanExporter.builder()
33             .setEndpoint(otlpEndpoint)
34             .build();
35
36         SdkTracerProvider sdkTracerProvider = SdkTracerProvider.builder()
37             .addSpanProcessor(BatchSpanProcessor.builder(spanExporter).build())
38             .setResource(resource)
39             .build();
40
41         return OpenTelemetrySdk.builder()
42             .setTracerProvider(sdkTracerProvider)
43             .setPropagators(ContextPropagators.create(W3CTraceContextPropagator.getInstance()))
44             .buildAndRegisterGlobal();
45     }

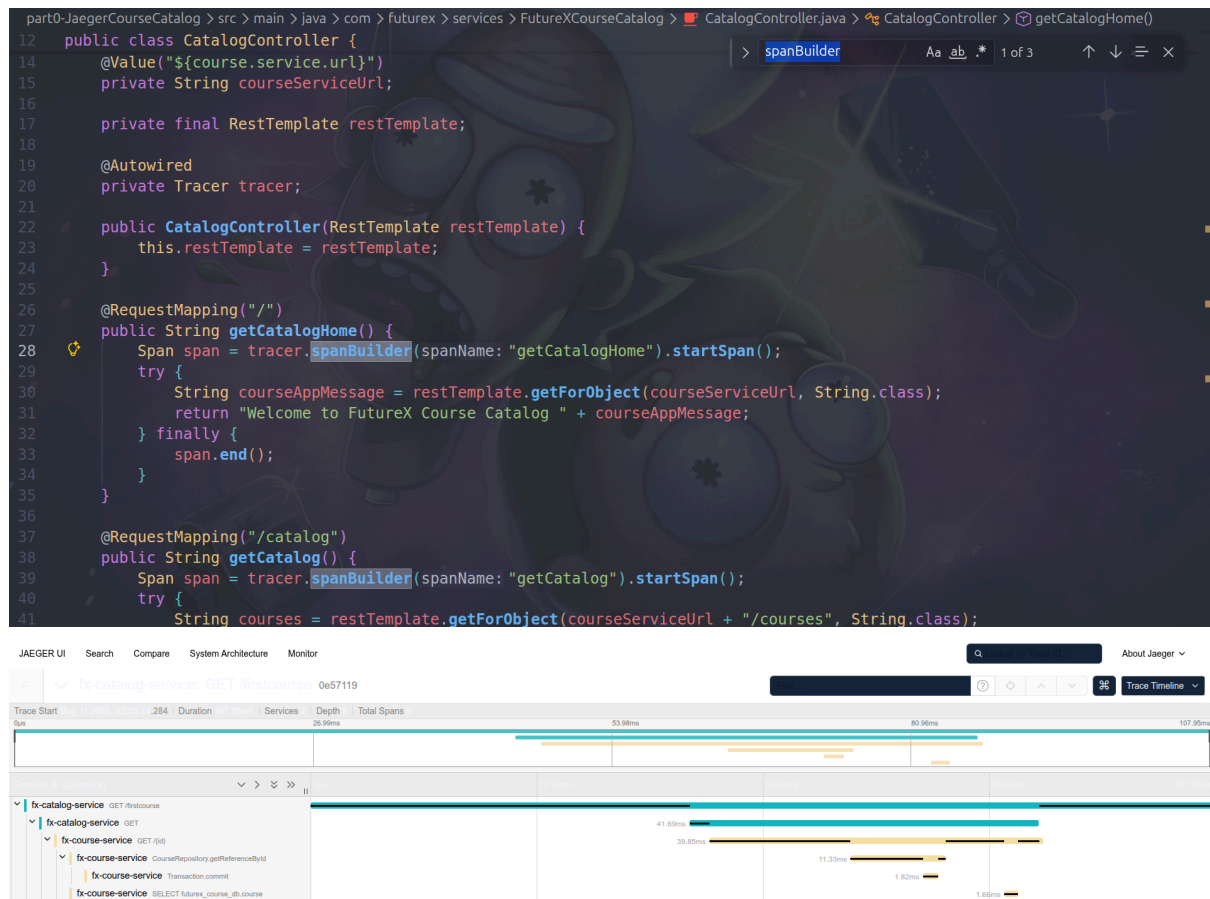
```

Se muestran los spans manuales dentro CourseController y CatalogController.

```

parte0-JaegerCourseApp > src > main > java > com > futurex > services > FutureXCourseApp > CourseController.java > CourseController > getCourseAppHome()
11 @RestController
12 public class CourseController {
13
14     @Autowired
15     private CourseRepository courseRepository;
16
17     @Autowired
18     private Tracer tracer;
19
20     @RequestMapping("/")
21     public String getCourseAppHome() {
22         Span span = tracer.spanBuilder("getCourseAppHome").startSpan();
23         try {
24             return "Course App Home";
25         } finally {
26             span.end();
27         }
28     }
29
30     @RequestMapping("/courses")
31     public List<Course> getCourses() {
32         Span span = tracer.spanBuilder("getCourses").startSpan();
33         try {
34             return courseRepository.findAll();
35         } finally {
36             span.end();
37         }
38     }
39 }

```



Parte 3 - Colector

Configuración del Colector


```

otel-collector-config.yaml
1  receivers:
2    otlp:
3      protocols:
4        grpc:
5        http:
6
7  processors:
8    batch:
9
10 exporters:
11   otlp:
12     endpoint: "jaeger:4317"
13     tls:
14       insecure: true
15
16 service:
17   pipelines:
18     traces:
19       receivers: [otlp]
20       processors: [batch]
21       exporters: [otlp]
22

```

```

parte0-JaegerCourseApp > src > main > resources > application.properties
1  # MySQL Database Configuration
2  spring.datasource.url=jdbc:mysql://localhost:3306/futurex_course_db?createDatabaseIfNotExist=true
3  spring.datasource.username=root
4  spring.datasource.password=techbankRootPsw
5  spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver
6
7  # JPA/Hibernate properties
8  spring.jpa.database-platform=org.hibernate.dialect.MySQL8Dialect
9  spring.jpa.show-sql=true
10 spring.jpa.hibernate.ddl-auto=update
11
12 # Application name and port
13 spring.application.name=fx-course-service
14 server.port=8001
15
16 # OpenTelemetry configuration
17 otel.exporter.otlp.endpoint=http://localhost:4319
18 otel.exporter.otlp.protocol=grpc
19 otel.metrics.exporter=none
20 otel.logs.exporter=none
21 otel.traces.exporter=otlp

```

Trazas

JAEGER UI

Search

Compare

System Architecture

Monitor

fx-catalog-service

getSpecificCourse

0e6296c

Trace Start

0.000000000000000000

Duration

0.000000000000000000

Services

Depth

3.67ms

Total Spans

11ms

14.68ms

fx-catalog-service

getSpecificCourse

Collector en Docker

```
torres@torres-Lenovo:~$ sudo docker ps
[sudo] contraseña para torres:
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
3a2df62482c1   otel/opentelemetry-collector:0.88.0 "/otelcol --config=/..." 2 hours ago    Up 2 hours    55678-55679/tcp, 0.0.0.0:4319->4317/tcp, [::]:4319->4317/tcp, 0.0.0.0:4320->4318/tcp, [::]:4320->4318/tcp
7a9de0b6c3bf   mysql:8.0                          "docker-entrypoint.s..." 2 hours ago    Up 2 hours    0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp
549f2b3b859c   jaegertracing/all-in-one:latest    "/go/bin/all-in-one-..." 2 hours ago    Up 2 hours    0.0.0.0:4317-4318->4317-4318/tcp, [::]:4317-4318->4317-4318/tcp, 9411/tcp, 0.0.0.0:14250->14250/tcp, [::]:14250->14250/tcp, 0.0.0.0:16686->16686/tcp, [::]:16686->16686/tcp, 14268/tcp
82f345d97f29   postgres:13                        "docker-entrypoint.s..." 3 weeks ago    Up 15 hours    0.0.0.0:5433->5432/tcp, [::]:5433->5432/tcp
a3a28ab7cf9b   dpape/pgadmin4                    "/entrypoint.sh"         5 weeks ago    Up 15 hours    443/tcp, 0.0.0.0:5050->80/tcp, [::]:5050->80/tcp
```